A universal bounded-degree polynomial test for separable camera-index re-scalings of quadrifocal determinants

Abstract

Let $n \geq 5$ and let $A^{(1)}, \ldots, A^{(n)} \in \mathbb{R}^{3\times 4}$ be Zariski-generic. From these matrices one forms, for each ordered camera quadruple $(\alpha, \beta, \gamma, \delta) \in [n]^4$, a tensor $Q^{(\alpha\beta\gamma\delta)} \in \mathbb{R}^{3\times 3\times 3\times 3}$ whose entries are 4×4 determinants of one row from each of the four matrices. Given a scalar array $\lambda \in \mathbb{R}^{n\times n\times n\times n}$ that is nonzero precisely on camera tuples that are not all identical, we consider the scaled collection $\left(\lambda_{\alpha\beta\gamma\delta} Q^{(\alpha\beta\gamma\delta)}\right)_{\alpha,\beta,\gamma,\delta}$. We construct, for each n, a polynomial map $\mathbf{F}$ of degree 5 (independent of n and of the cameras) that vanishes on this scaled collection if and only if λ factors as $\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta$ for all non-identical $(\alpha, \beta, \gamma, \delta)$ with $u, v, w, x \in (\mathbb{R}^*)^n$.

1 Problem and reformulation

Fix $n \geq 5$. For $\alpha \in [n] := \{1, \ldots, n\}$ let $A^{(\alpha)} \in \mathbb{R}^{3\times 4}$. For $(\alpha, \beta, \gamma, \delta) \in [n]^4$ define $Q^{(\alpha\beta\gamma\delta)} \in \mathbb{R}^{3\times 3\times 3\times 3}$ by

$$Q^{(\alpha\beta\gamma\delta)}_{ijkl} := \det \begin{bmatrix} A^{(\alpha)}(i,:) \\ A^{(\beta)}(j,:) \\ A^{(\gamma)}(k,:) \\ A^{(\delta)}(\ell,:) \end{bmatrix}, \qquad 1 \leq i, j, k, \ell \leq 3.$$

We are given $\lambda \in \mathbb{R}^{n\times n\times n\times n}$ with

$$\lambda_{\alpha\beta\gamma\delta} \neq 0 \iff (\alpha, \beta, \gamma, \delta) \text{ are not all identical,}$$

and we want a *universal* bounded-degree polynomial condition on the scaled family $\left(\lambda_{\alpha\beta\gamma\delta} Q^{(\alpha\beta\gamma\delta)}\right)$ that holds if and only if λ is *separable* in the sense that there exist $u, v, w, x \in (\mathbb{R}^*)^n$ with

$$\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta \quad \text{for all } (\alpha, \beta, \gamma, \delta) \text{ that are not all identical.} \tag{1}$$

It is convenient to bundle the n^4 tensors into a single 4-way tensor of side length $3n$.

1.1 Indexing

Let

$$m := 3n, \qquad I := [n] \times [3] = \{(\alpha, i) : \alpha \in [n],\, i \in [3]\}.$$

We identify I with $\{1, \ldots, m\}$ only for the purpose of matrix minors; conceptually we keep the pair notation. For $a = (\alpha, i) \in I$ define the row vector

$$r_a := A^{(\alpha)}(i,:) \in \mathbb{R}^{1\times 4}.$$

Let $M \in \mathbb{R}^{m\times 4}$ be the matrix with rows $(r_a)_{a\in I}$, i.e. M is the vertical stacking of all $A^{(\alpha)}$.

Define the *determinant tensor* $P \in \mathbb{R}^{I\times I\times I\times I}$ by

$$P_{abcd} := \det \begin{bmatrix} r_a \\ r_b \\ r_c \\ r_d \end{bmatrix} \qquad (a,b,c,d \in I). \tag{2}$$

Then, by construction,

$$Q^{(\alpha\beta\gamma\delta)}_{ijkl} = P_{(\alpha,i)\,(\beta,j)\,(\gamma,k)\,(\delta,\ell)}.$$

Now extend λ to a block-constant 4-tensor $L \in \mathbb{R}^{I\times I\times I\times I}$ by

$$L_{(\alpha,i)\,(\beta,j)\,(\gamma,k)\,(\delta,\ell)} := \lambda_{\alpha\beta\gamma\delta}.$$

The scaled data are entries of the Hadamard product tensor

$$T := L \odot P, \qquad T_{(\alpha,i)\,(\beta,j)\,(\gamma,k)\,(\delta,\ell)} = \lambda_{\alpha\beta\gamma\delta}\, Q^{(\alpha\beta\gamma\delta)}_{ijkl}.$$

Thus the question becomes: construct a universal bounded-degree polynomial map $\mathbf{F}$ in the entries of T whose vanishing is equivalent (for generic $A^{(\alpha)}$ and admissible λ) to the separability (1).

2 A candidate polynomial map via Tucker rank

For a 4-way tensor $X \in \mathbb{R}^{m\times m\times m\times m}$ and $p \in \{1,2,3,4\}$, let $X_{(p)}$ denote the *mode-p unfolding* (matricization), i.e. the $m \times m^3$ matrix obtained by using the p-th index as the row index and the remaining three indices as a multi-index for columns. It is standard that

$$\operatorname{rank}(X_{(p)}) \le 4 \quad \Longleftrightarrow \quad \text{all } 5\times 5 \text{ minors of } X_{(p)} \text{ vanish.}$$

Definition 1 (The polynomial map $\mathbf{F}_n$)**.** For each n (hence $m = 3n$), define $\mathbf{F}_n : \mathbb{R}^{m^4} \to \mathbb{R}^{N_n}$ by

$$\mathbf{F}_n(X) := \Big(\det\big(X_{(p)}[R,C]\big)\Big)_{\substack{p\in\{1,2,3,4\}\\ R\subseteq[m],\,|R|=5\\ C\subseteq[m^3],\,|C|=5}},$$

i.e. $\mathbf{F}_n$ outputs *all* 5×5 minors of the four unfoldings. Here $X_{(p)}[R,C]$ is the 5×5 submatrix with row set R and column set C.

Each coordinate of $\mathbf{F}_n$ is a determinant of a 5×5 matrix of entries of X, hence is a polynomial of degree 5, independent of n and of $A^{(\alpha)}$. The output dimension N_n is enormous but finite; this is permitted.

We will prove that, for Zariski-generic $(A^{(1)},\dots,A^{(n)})$, the vanishing of $\mathbf{F}_n(T)$ characterizes separability of λ in the sense of (1).

3 Generic structure of the determinant tensor

We work over $\mathbb{C}$ for the algebraic-geometric arguments; Zariski-genericity over $\mathbb{R}$ implies the same properties after complexification, and the final statement descends back to $\mathbb{R}$. So set $\Bbbk := \mathbb{C}$ for the proofs below.

3.1 Rank-4 structure of unfoldings

Let $M \in \Bbbk^{m\times 4}$ be the stacked matrix and let P be defined by (2). Write the columns of M as $M = [\, m_1 \; m_2 \; m_3 \; m_4 \,]$ with $m_s \in \Bbbk^m$.

Lemma 1 (Mode ranks of P)**.** *If* $\operatorname{rank}(M) = 4$*, then for each* $p \in \{1, 2, 3, 4\}$ *we have* $\operatorname{rank}(P_{(p)}) = 4$.

Proof. We prove the claim for $p = 1$; the other modes follow by symmetry of the determinant in the four arguments. Fix $(b, c, d) \in I^3$ and consider the column vector $P_{:,b,c,d} \in \Bbbk^m$ with entries $P_{a,b,c,d} = \det(r_a, r_b, r_c, r_d)$. Expanding the 4×4 determinant along the first row yields a cofactor vector $n(b, c, d) \in \Bbbk^4$ such that

$$\det(r_a, r_b, r_c, r_d) = r_a\, n(b, c, d) \quad \text{for all } a \in I,$$

where we view $r_a \in \Bbbk^{1\times 4}$ as a row vector and $n(b, c, d)$ as a column vector. Therefore

$$P_{:,b,c,d} = M\, n(b, c, d) \in \operatorname{Col}(M).$$

Hence $\operatorname{Col}(P_{(1)}) \subseteq \operatorname{Col}(M)$, so $\operatorname{rank}(P_{(1)}) \le 4$.

To see $\operatorname{rank}(P_{(1)}) = 4$ for generic M, it suffices to exhibit some (b, c, d) for which $n(b, c, d)$ span $\Bbbk^4$, because then the columns $Mn(b, c, d)$ span $\operatorname{Col}(M)$. For Zariski-generic M of rank 4, the set of cofactor vectors $\{n(b, c, d)\}$ spans $\Bbbk^4$, so $\operatorname{Col}(P_{(1)}) = \operatorname{Col}(M)$ and $\operatorname{rank}(P_{(1)}) = 4$. □

3.2 A spanning lemma for camera-triples

For our main argument we need that, for many fixed choices of the *camera indices* in three modes, the corresponding block of columns already spans $\operatorname{Col}(M)$.

For $\beta, \gamma, \delta \in [n]$ define the index subsets

$$I_\beta := \{(\beta, i) : i \in [3]\} \subset I,$$

and let $P_{(1)}^{\beta\gamma\delta}$ be the submatrix of $P_{(1)}$ consisting of columns indexed by triples $(b, c, d) \in I_\beta \times I_\gamma \times I_\delta$.

Lemma 2 (Generic spanning for non-constant camera triples)**.** *Assume* $\operatorname{rank}(M) = 4$ *and each* $A^{(\alpha)}$ *has rank* 3*. There exists a Zariski-open dense subset* $\mathcal{U} \subset (\Bbbk^{3\times 4})^n$ *such that, for all* $(A^{(1)}, \ldots, A^{(n)}) \in \mathcal{U}$ *and all ordered triples* $(\beta, \gamma, \delta) \in [n]^3$ *that are* not all equal*, the column space of* $P_{(1)}^{\beta\gamma\delta}$ *equals* $\operatorname{Col}(M)$.

Proof. Fix a triple (β, γ, δ) with not all equal. As in the proof of Lemma 1, columns of $P_{(1)}^{\beta\gamma\delta}$ are of the form $M\, n(b, c, d)$ where $(b, c, d) \in I_\beta \times I_\gamma \times I_\delta$. Thus $\operatorname{Col}(P_{(1)}^{\beta\gamma\delta}) = \operatorname{Col}(M \cdot \mathcal{N})$ where $\mathcal{N}$ is the span of the 27 vectors $n(b, c, d) \in \Bbbk^4$ arising from these triples of rows.

Since $\operatorname{rank}(M) = 4$, we have $\operatorname{Col}(P_{(1)}^{\beta\gamma\delta}) = \operatorname{Col}(M)$ if and only if $\mathcal{N} = \Bbbk^4$. The condition $\mathcal{N} = \Bbbk^4$ is Zariski-open: it is equivalent to the nonvanishing of at least one 4×4 determinant formed by four of the vectors $n(b, c, d)$.

It remains to show this open condition is nonempty for the fixed triple (β, γ, δ). We exhibit one explicit choice of three 3×4 matrices realizing $\mathcal{N} = \Bbbk^4$.

Let $e_1, \ldots, e_4$ be the standard basis of $\Bbbk^4$ (row vectors). If β, γ, δ are all distinct, set

$$A^{(\beta)} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}, \quad A^{(\gamma)} = \begin{bmatrix} e_4 \\ e_1 \\ e_2 \end{bmatrix}, \quad A^{(\delta)} = \begin{bmatrix} e_3 \\ e_4 \\ e_1 \end{bmatrix}.$$

Then one checks (by direct wedge/Hodge-dual computation, or by cofactor expansion) that among the 27 choices (b, c, d) there are four giving cofactor vectors $n(b, c, d)$ equal to e_1, e_2, e_3, e_4, hence spanning $\Bbbk^4$.

If exactly two of β, γ, δ are equal, say $\beta = \gamma \neq \delta$, use the same assignment but interpret the two occurrences of β as choosing two (possibly distinct) rows from $A^{(\beta)}$; again the 27 possible choices include four triples producing four linearly independent cofactor vectors (for example by selecting two distinct rows from $A^{(\beta)}$ together with suitable rows from $A^{(\delta)}$).

Therefore, for each fixed triple (β, γ, δ) with not all equal, the condition $\mathcal{N} = \Bbbk^4$ is satisfied by some choice of the involved camera matrices, hence the corresponding Zariski-open set is nonempty. Because there are only finitely many such triples for fixed n, the intersection over all of them is again a nonempty Zariski-open set $\mathcal{U}$ on which the claim holds simultaneously. □

3.3 A stabilizer lemma for block-diagonal scalings

Let $\mathcal{D} \subset \mathrm{GL}_m(\Bbbk)$ be the torus of block-diagonal matrices

$$D(t) := \mathrm{diag}\big(t_1 \mathrm{Id}_3, \ldots, t_n \mathrm{Id}_3\big) \qquad (t_1, \ldots, t_n \in \Bbbk^*),$$

acting on $\Bbbk^m$ by scaling each camera-block of three coordinates.

Lemma 3 (Generic stabilizer is scalar)**.** *For Zariski-generic $A^{(1)}, \ldots, A^{(n)}$, the only $D(t) \in \mathcal{D}$ satisfying $D(t)\,\mathrm{Col}(M) = \mathrm{Col}(M)$ are the scalar matrices $t_1 = \cdots = t_n$.*

Proof. Suppose $D(t)\,\mathrm{Col}(M) \subseteq \mathrm{Col}(M)$. Then there exists a 4×4 matrix S such that

$$D(t)\, M = M\, S. \tag{3}$$

Now restrict (3) to the three rows corresponding to camera α. If $M_\alpha = A^{(\alpha)} \in \Bbbk^{3\times 4}$ denotes that block, then (3) implies

$$t_\alpha\, M_\alpha = M_\alpha\, S.$$

Equivalently, for every row vector r in the row space $U_\alpha := \mathrm{Row}(M_\alpha) \subset \Bbbk^{1\times 4}$, we have

$$r\, S = t_\alpha\, r.$$

Thus U_α is contained in the left-eigenspace of S with eigenvalue t_α. Since M_α has rank 3 generically, $\dim U_\alpha = 3$ for each α.

If $t_\alpha \neq t_\beta$, then U_α and U_β would lie in left-eigenspaces of distinct eigenvalues of S, hence would intersect trivially. But two 3-dimensional subspaces of a 4-dimensional space must intersect in dimension at least 2:

$$\dim(U_\alpha \cap U_\beta) \geq \dim U_\alpha + \dim U_\beta - 4 = 2.$$

Therefore $t_\alpha = t_\beta$ for all α, β, i.e. $D(t)$ is scalar. □

4 Rank-4 unfoldings force separable camera-index scaling

Let λ be as in the problem statement (nonzero on camera tuples not all identical) and set $T = L \odot P$.

For $(\beta, \gamma, \delta) \in [n]^3$ define the block-diagonal matrix on the first mode

$$D^{(1)}_{\beta\gamma\delta} := \mathrm{diag}\big(\lambda_{1\beta\gamma\delta} \mathrm{Id}_3, \ldots, \lambda_{n\beta\gamma\delta} \mathrm{Id}_3\big) \in \Bbbk^{m\times m}.$$

Then for any $b \in I_\beta$, $c \in I_\gamma$, $d \in I_\delta$ we have

$$T_{:,b,c,d} = D^{(1)}_{\beta\gamma\delta}\, P_{:,b,c,d}.$$

Therefore the submatrix $T^{\beta\gamma\delta}_{(1)}$ (columns of $T_{(1)}$ with camera-triple (β,γ,δ)) satisfies

$$\mathrm{Col}(T^{\beta\gamma\delta}_{(1)}) = D^{(1)}_{\beta\gamma\delta}\,\mathrm{Col}(P^{\beta\gamma\delta}_{(1)}).$$

Proposition 1 (Mode-1 rank-4 implies separation in the first camera index)**.** *Assume* $(A^{(1)},\dots,A^{(n)}) \in \mathcal{U}$ *as in Lemma 2. If* $\mathrm{rank}(T_{(1)}) \le 4$*, then there exist* $u \in (\Bbbk^*)^n$ *and* $\mu \in \Bbbk^{n\times n\times n}$ *such that*

$$\lambda_{\alpha\beta\gamma\delta} = u_\alpha\,\mu_{\beta\gamma\delta} \quad \textit{for all } \alpha \in [n] \textit{ and all } (\beta,\gamma,\delta) \textit{ not all equal.} \tag{4}$$

Proof. Fix any camera-triple (β,γ,δ) that is not all equal. By Lemma 2, $\mathrm{Col}(P^{\beta\gamma\delta}_{(1)}) = \mathrm{Col}(M)$. Hence

$$\mathrm{Col}(T^{\beta\gamma\delta}_{(1)}) = D^{(1)}_{\beta\gamma\delta}\,\mathrm{Col}(M).$$

Since $\mathrm{rank}(T_{(1)}) \le 4$, the column space $\mathrm{Col}(T_{(1)})$ has dimension at most 4. But for our admissible λ, and for any (β,γ,δ) not all equal, each scalar $\lambda_{\alpha\beta\gamma\delta}$ is nonzero for all α (because the full quadruple $(\alpha,\beta,\gamma,\delta)$ is never all identical when (β,γ,δ) is not all equal). Thus $D^{(1)}_{\beta\gamma\delta}$ is invertible, so $\dim(D^{(1)}_{\beta\gamma\delta}\,\mathrm{Col}(M)) = \dim\mathrm{Col}(M) = 4$ (genericity gives $\mathrm{rank}(M) = 4$). Therefore $\mathrm{Col}(T^{\beta\gamma\delta}_{(1)})$ is a 4-dimensional subspace contained in $\mathrm{Col}(T_{(1)})$, forcing

$$\mathrm{Col}(T_{(1)}) = D^{(1)}_{\beta\gamma\delta}\,\mathrm{Col}(M) \qquad \text{for all } (\beta,\gamma,\delta) \text{ not all equal.}$$

In particular, for any two such triples (β,γ,δ) and (β',γ',δ') we have

$$D^{(1)}_{\beta\gamma\delta}\,\mathrm{Col}(M) = D^{(1)}_{\beta'\gamma'\delta'}\,\mathrm{Col}(M).$$

Thus

$$\big(D^{(1)}_{\beta'\gamma'\delta'}\big)^{-1} D^{(1)}_{\beta\gamma\delta}\,\mathrm{Col}(M) = \mathrm{Col}(M),$$

and Lemma 3 implies $(D^{(1)}_{\beta'\gamma'\delta'})^{-1} D^{(1)}_{\beta\gamma\delta}$ is scalar. Hence there exist scalars $\mu_{\beta\gamma\delta} \in \Bbbk^*$ and a fixed block-diagonal $D_0 \in \mathcal{D}$ such that

$$D^{(1)}_{\beta\gamma\delta} = \mu_{\beta\gamma\delta}\, D_0 \quad \text{for all } (\beta,\gamma,\delta) \text{ not all equal.}$$

Let u_α be the block scalar of D_0 on camera α. Then the (α,α) block equality gives

$$\lambda_{\alpha\beta\gamma\delta} = u_\alpha\,\mu_{\beta\gamma\delta},$$

which is (4). □

By symmetry (permuting the roles of the four modes), we also obtain analogous separations from ranks of the other unfoldings.

Corollary 1. *Under the same genericity assumptions as above, if* $\mathrm{rank}(T_{(p)}) \le 4$ *for* $p = 1,2,3$*, then there exist* $u,v,w \in (\Bbbk^*)^n$ *and arrays* μ,ν,ξ *such that*

$$\lambda_{\alpha\beta\gamma\delta} = u_\alpha \mu_{\beta\gamma\delta} \quad \textit{whenever } (\beta,\gamma,\delta) \textit{ not all equal,}$$

$$\lambda_{\alpha\beta\gamma\delta} = v_\beta \nu_{\alpha\gamma\delta} \quad \textit{whenever } (\alpha,\gamma,\delta) \textit{ not all equal,}$$

$$\lambda_{\alpha\beta\gamma\delta} = w_\gamma \xi_{\alpha\beta\delta} \quad \textit{whenever } (\alpha,\beta,\delta) \textit{ not all equal.}$$

5 From mode-wise separations to full rank-one factorization

We now show that the three separations above imply full separability $\lambda = u \otimes v \otimes w \otimes x$ on all camera quadruples that are not all identical.

Lemma 4 (Two separations imply a third)**.** *Assume $\lambda_{\alpha\beta\gamma\delta} \neq 0$ for all $(\alpha, \beta, \gamma, \delta)$ not all identical. Suppose there exist $u, v \in (\Bbbk^*)^n$ and arrays μ, ν such that*

$$\lambda_{\alpha\beta\gamma\delta} = u_\alpha \mu_{\beta\gamma\delta} \quad \textit{whenever } (\beta, \gamma, \delta) \textit{ not all equal,}$$

$$\lambda_{\alpha\beta\gamma\delta} = v_\beta \nu_{\alpha\gamma\delta} \quad \textit{whenever } (\alpha, \gamma, \delta) \textit{ not all equal.}$$

Then there exists an array $\kappa \in \Bbbk^{n\times n}$ such that

$$\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta \kappa_{\gamma\delta} \quad \textit{for all } (\alpha, \beta, \gamma, \delta) \textit{ not all identical.} \tag{5}$$

Proof. Fix $(\gamma, \delta) \in [n]^2$. Choose some $\beta_0 \in [n]$ such that $(\beta_0, \gamma, \delta)$ is not all equal. This is always possible: if $\gamma \neq \delta$ take any β_0, and if $\gamma = \delta$ take $\beta_0 \neq \gamma$.

For any $\alpha \in [n]$ with (α, γ, δ) not all equal (equivalently, not $\alpha = \gamma = \delta$), both assumed formulas apply to the tuple $(\alpha, \beta_0, \gamma, \delta)$, giving

$$u_\alpha \, \mu_{\beta_0\gamma\delta} = \lambda_{\alpha\beta_0\gamma\delta} = v_{\beta_0} \, \nu_{\alpha\gamma\delta}.$$

Hence for all such α,

$$\nu_{\alpha\gamma\delta} = u_\alpha \, \kappa_{\gamma\delta} \quad \text{where} \quad \kappa_{\gamma\delta} := \frac{\mu_{\beta_0\gamma\delta}}{v_{\beta_0}} \in \Bbbk^*.$$

Now let $(\alpha, \beta, \gamma, \delta)$ be any camera quadruple not all identical.

Case 1: (α, γ, δ) not all equal. Then

$$\lambda_{\alpha\beta\gamma\delta} = v_\beta \nu_{\alpha\gamma\delta} = v_\beta \, u_\alpha \, \kappa_{\gamma\delta},$$

which is (5).

Case 2: (α, γ, δ) all equal. Then $\alpha = \gamma = \delta$ and, since the quadruple is not all identical, necessarily $\beta \neq \alpha$. In this case $(\beta, \gamma, \delta) = (\beta, \alpha, \alpha)$ is not all equal, so the first separation applies: $\lambda_{\alpha\beta\alpha\alpha} = u_\alpha \mu_{\beta\alpha\alpha}$. Choose some $\alpha' \neq \alpha$ (possible because $n \geq 2$, and in our setting $n \geq 5$). Then $(\alpha', \alpha, \alpha)$ is not all equal, so Case 1 gives

$$\lambda_{\alpha'\beta\alpha\alpha} = u_{\alpha'} v_\beta \kappa_{\alpha\alpha}.$$

But also (since $\beta \neq \alpha$) the first separation gives $\lambda_{\alpha'\beta\alpha\alpha} = u_{\alpha'} \mu_{\beta\alpha\alpha}$. Cancelling $u_{\alpha'} \neq 0$ yields $\mu_{\beta\alpha\alpha} = v_\beta \kappa_{\alpha\alpha}$, and hence

$$\lambda_{\alpha\beta\alpha\alpha} = u_\alpha \mu_{\beta\alpha\alpha} = u_\alpha v_\beta \kappa_{\alpha\alpha}.$$

Thus (5) holds also in Case 2. □

Lemma 5 (Separating κ)**.** *Assume (5) holds with $u, v \in (\Bbbk^*)^n$ and some $\kappa \in \Bbbk^{n\times n}$. Suppose also there exist $w \in (\Bbbk^*)^n$ and an array ξ such that*

$$\lambda_{\alpha\beta\gamma\delta} = w_\gamma \, \xi_{\alpha\beta\delta} \quad \textit{whenever } (\alpha, \beta, \delta) \textit{ not all equal.}$$

Then there exists $x \in (\Bbbk^*)^n$ such that

$$\kappa_{\gamma\delta} = w_\gamma x_\delta \qquad \textit{for all } \gamma, \delta \in [n].$$

Proof. Fix $\delta \in [n]$. Choose some pair (α_0, β_0) with $(\alpha_0, \beta_0, \delta)$ not all equal; this is always possible because we can take $\alpha_0 \neq \delta$ and any β_0. Then for all $\gamma \in [n]$, the tuple $(\alpha_0, \beta_0, \gamma, \delta)$ is not all identical, and we have both (5) and the assumed γ-separation:

$$u_{\alpha_0} v_{\beta_0} \kappa_{\gamma\delta} = \lambda_{\alpha_0\beta_0\gamma\delta} = w_\gamma \, \xi_{\alpha_0\beta_0\delta}.$$

The right-hand side is a scalar multiple of w_γ with a coefficient independent of γ, while the left-hand side is a scalar multiple of $\kappa_{\gamma\delta}$ with a coefficient independent of γ. Since $u_{\alpha_0} v_{\beta_0} \neq 0$ and $\xi_{\alpha_0\beta_0\delta} \neq 0$ (by nonvanishing of λ on non-identical tuples), it follows that the vector $(\kappa_{\gamma\delta})_\gamma$ is proportional to $(w_\gamma)_\gamma$. Thus there exists $x_\delta \in \Bbbk^*$ such that $\kappa_{\gamma\delta} = w_\gamma x_\delta$ for all γ. Doing this for each δ defines $x \in (\Bbbk^*)^n$ as desired. □

Combining Lemmas 4 and 5 yields the desired rank-one factorization.

Theorem 1 (Existence of a universal bounded-degree polynomial test)**.** *Let $n \geq 5$. There exists a polynomial map $\mathbf{F}_n : \mathbb{R}^{81n^4} \to \mathbb{R}^{N_n}$ satisfying:*

1. *$\mathbf{F}_n$ depends only on n (not on $A^{(1)}, \ldots, A^{(n)}$);*
2. *every coordinate polynomial of $\mathbf{F}_n$ has degree 5 (independent of n);*
3. *for Zariski-generic $A^{(1)}, \ldots, A^{(n)} \in \mathbb{R}^{3\times 4}$ and for any $\lambda \in \mathbb{R}^{n\times n\times n\times n}$ with $\lambda_{\alpha\beta\gamma\delta} \neq 0$ iff $(\alpha, \beta, \gamma, \delta)$ are not all identical,*

$$\mathbf{F}_n\big(\lambda_{\alpha\beta\gamma\delta} Q^{(\alpha\beta\gamma\delta)} : \alpha, \beta, \gamma, \delta \in [n]\big) = 0$$

if and only if there exist $u, v, w, x \in (\mathbb{R}^*)^n$ such that

$$\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta \qquad \text{for all } (\alpha, \beta, \gamma, \delta) \text{ not all identical.}$$

Proof. Define $\mathbf{F}_n$ as in Definition 1, i.e. all 5×5 minors of the four unfoldings.

(⇒) Suppose $\mathbf{F}_n(T) = 0$ for $T = L \odot P$. Then every mode unfolding $T_{(p)}$ has rank ≤ 4. Apply Proposition 1 in modes $p = 1, 2, 3$ (which is valid for Zariski-generic cameras by Lemmas 2 and 3) to obtain the three partial separations in Corollary 1. Then Lemma 4 yields $\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta \kappa_{\gamma\delta}$ for all non-identical camera quadruples. Finally Lemma 5 yields $\kappa_{\gamma\delta} = w_\gamma x_\delta$, hence $\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta$ for all non-identical tuples, as required.

(⇐) Conversely, suppose there exist $u, v, w, x \in (\mathbb{R}^*)^n$ such that $\lambda_{\alpha\beta\gamma\delta} = u_\alpha v_\beta w_\gamma x_\delta$ for all non-identical camera tuples. Extend this definition to *all* tuples by setting $\tilde{\lambda}_{\alpha\beta\gamma\delta} := u_\alpha v_\beta w_\gamma x_\delta$ for every $(\alpha, \beta, \gamma, \delta)$. Because $Q^{(\alpha\alpha\alpha\alpha)} \equiv 0$ (any 4×4 determinant with four rows from a 3×4 matrix has repeated rows), we have

$$\lambda_{\alpha\beta\gamma\delta} Q^{(\alpha\beta\gamma\delta)} = \tilde{\lambda}_{\alpha\beta\gamma\delta} Q^{(\alpha\beta\gamma\delta)} \quad \text{for all } (\alpha, \beta, \gamma, \delta).$$

Hence $T = L \odot P$ equals the tensor obtained from P by mode-wise multiplication with block-diagonal matrices D_u, D_v, D_w, D_x that scale camera-block α by $u_\alpha, v_\alpha, w_\alpha, x_\alpha$ in modes $1, 2, 3, 4$ respectively. Mode-wise multiplication by invertible matrices does not increase unfolding ranks, and since P has mode ranks 4 (Lemma 1), so does T. Therefore all 5×5 minors of each unfolding vanish and $\mathbf{F}_n(T) = 0$.

This proves the equivalence. □

Remark 1. The construction of $\mathbf{F}_n$ is completely explicit and universal: it is the (huge) list of 5×5 minors of the four unfoldings of the stacked tensor T. The degree is exactly 5 independent of n.